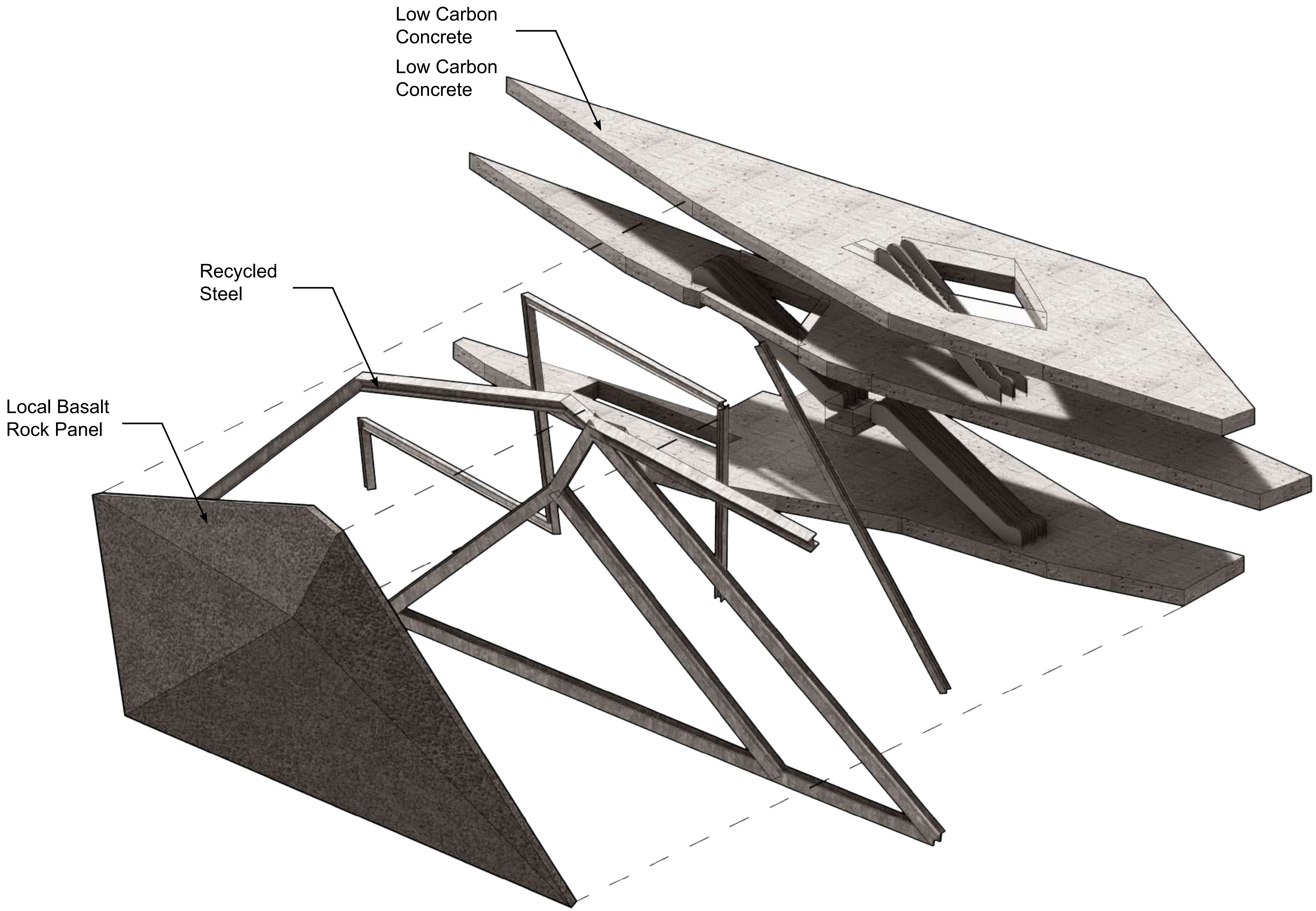
E4
Material Research

The project integrates material strategies that reinforce both the architectural concept and environmental responsibility. First, the use of locally sourced black rock establishes a direct relationship between the building and its geographic context. Incorporating regional stone strengthens the project's design synthesis by grounding the architecture in the material identity of its site. At the same time, sourcing stone locally reduces the environmental impact associated with long distance material transport and avoids the need to synthetically reproduce similar materials through energy intensive industrial processes.

- Concrete can account for **30–50% of total** building embodied carbon
- **300–400 kg CO2 e/m³** → Conventional Concrete
- **150–250 kg CO2 e/m³** → Low Carbon Mix
- Up to **50% CO2 Reduction** (SCM substitution)
- Cement = **~90% of concrete emissions**
- **30–50%** Cement Replacement Strategy
- Concrete = up to **50% of building embodied carbon**
- **Transport can add 5–15% additional** CO2 depending on distance



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